CS 5/7320
Artificial Intelligence

Knowledge-Based Agents
AIMA Chapters 7-9

Slides by Michael Hahsler

based on slides by Svetlana Lazepnik
with figures from the AIMA textbook

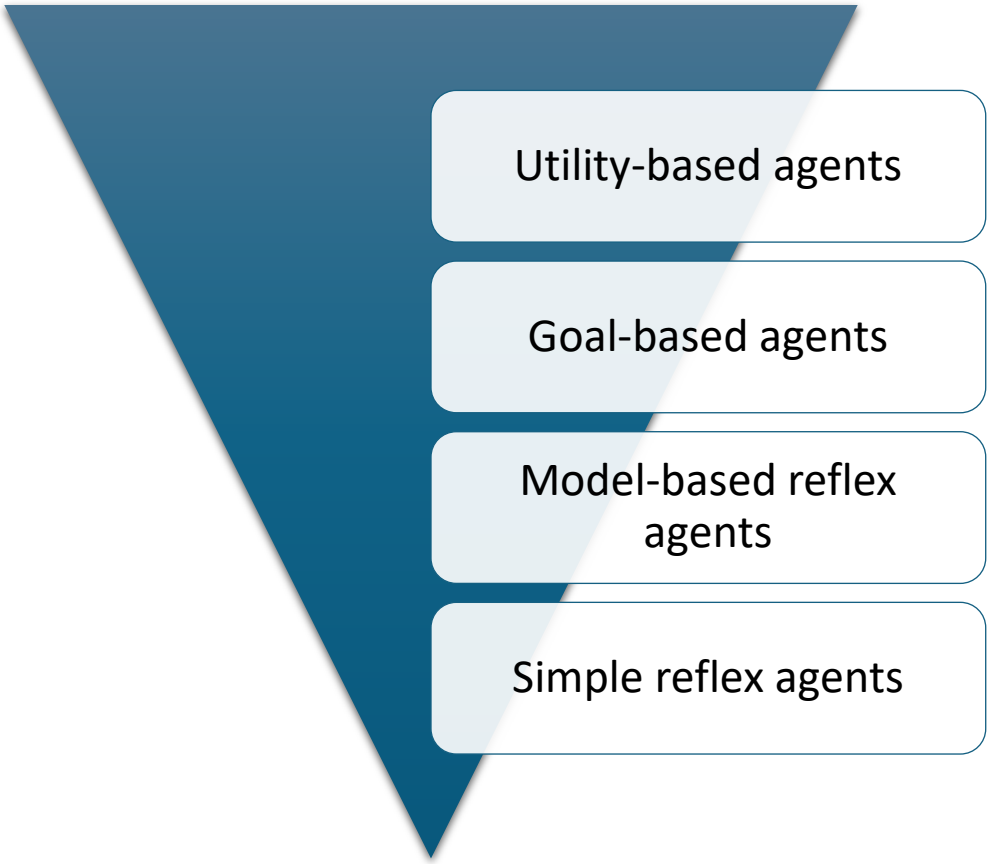


This work is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

Image: "Exercise Plays Vital Role Maintaining Brain Health" by A Health Blog

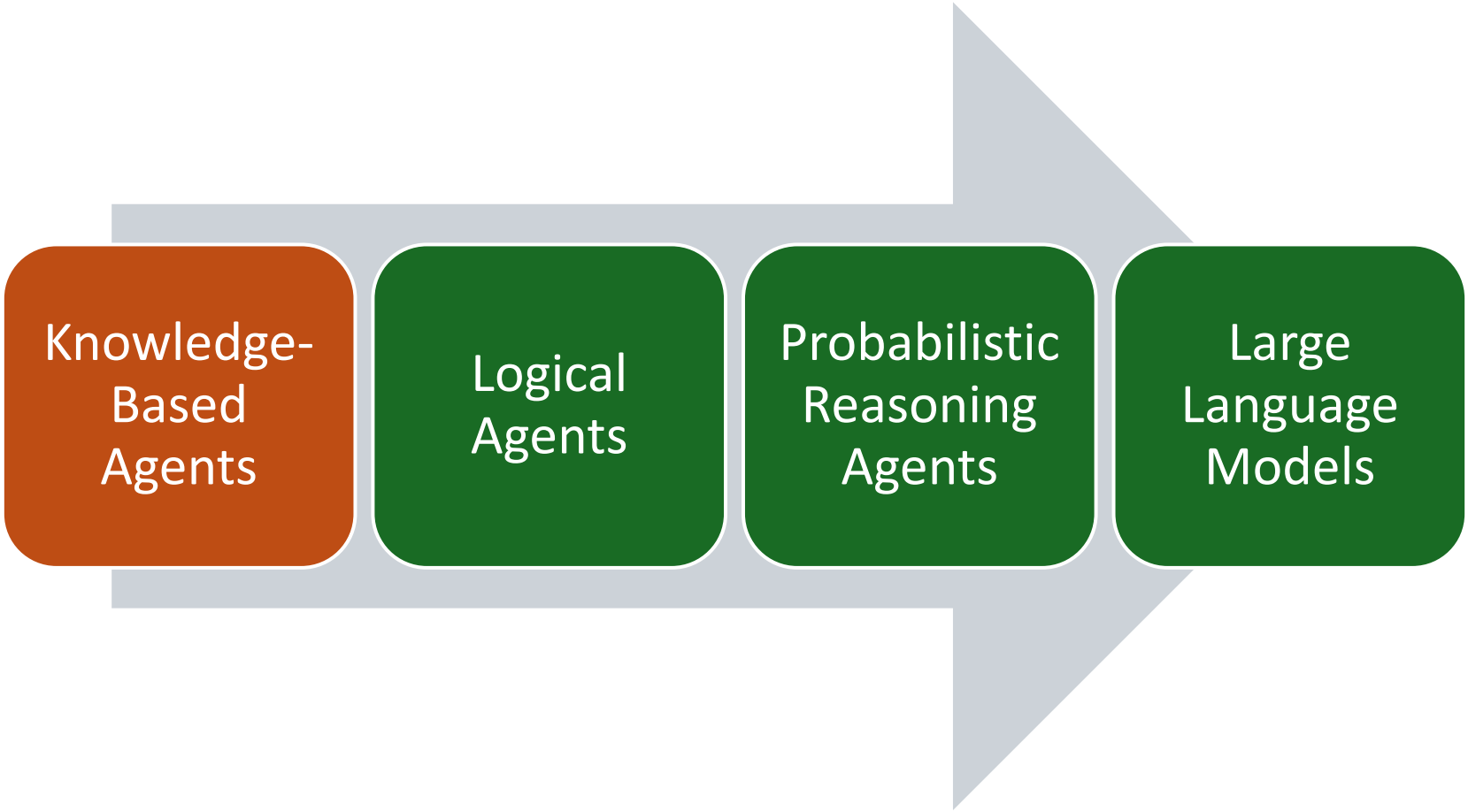
Online Material

Agent Types So Far

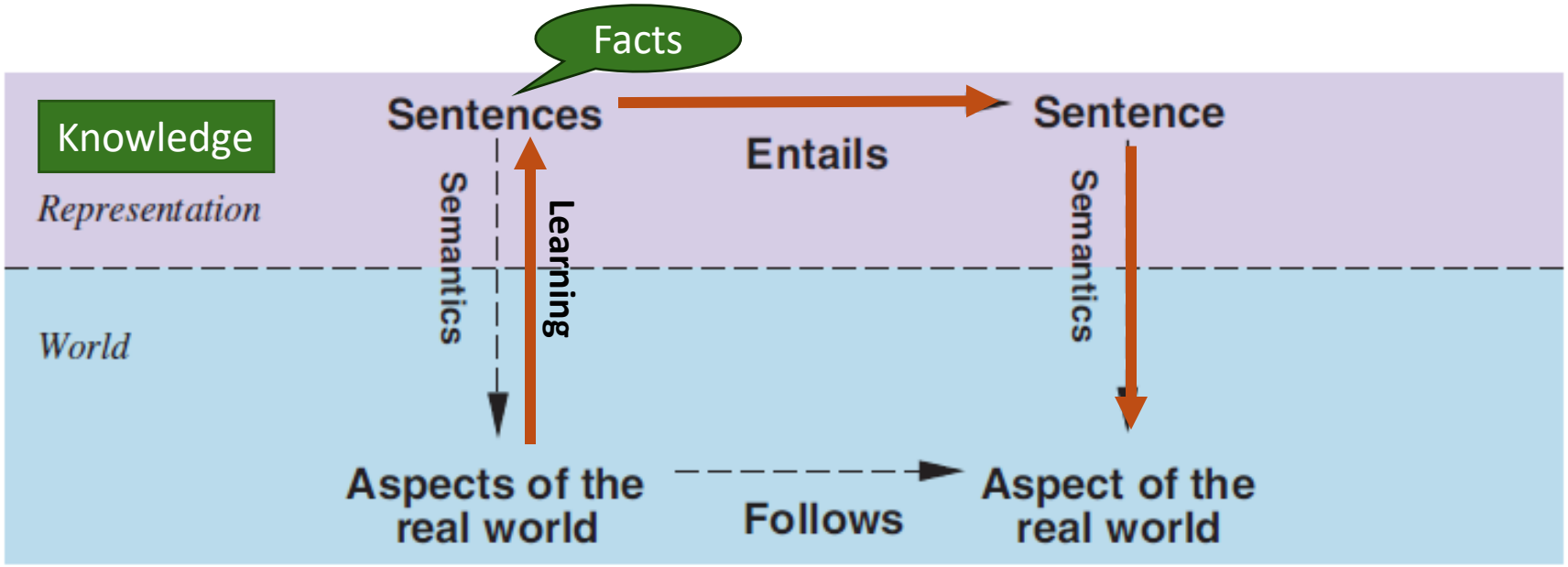


We will introduce a different class of agents that relies on reasoning using knowledge.

Outline

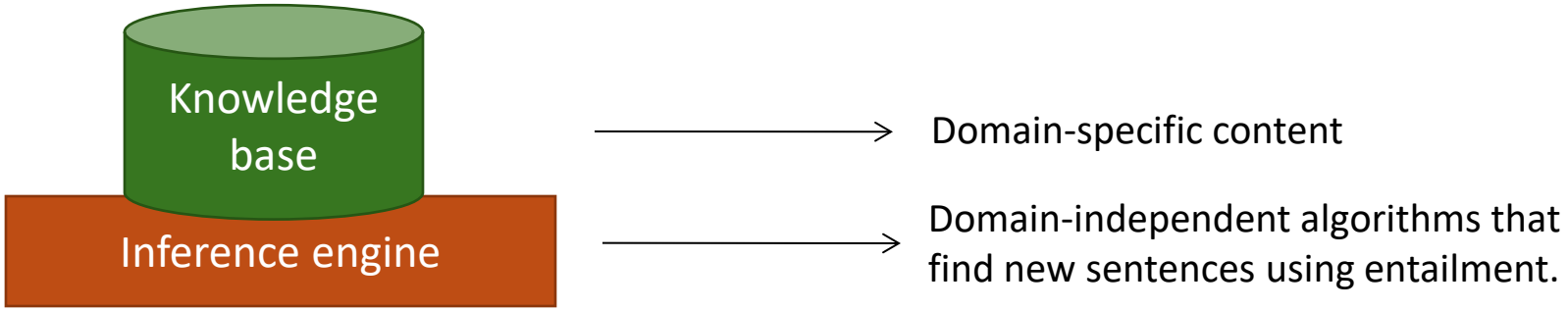

Knowledge-Based Agents
Logical Agents
Probabilistic Reasoning Agents
Large Language Models

Reality vs. Knowledge Representation



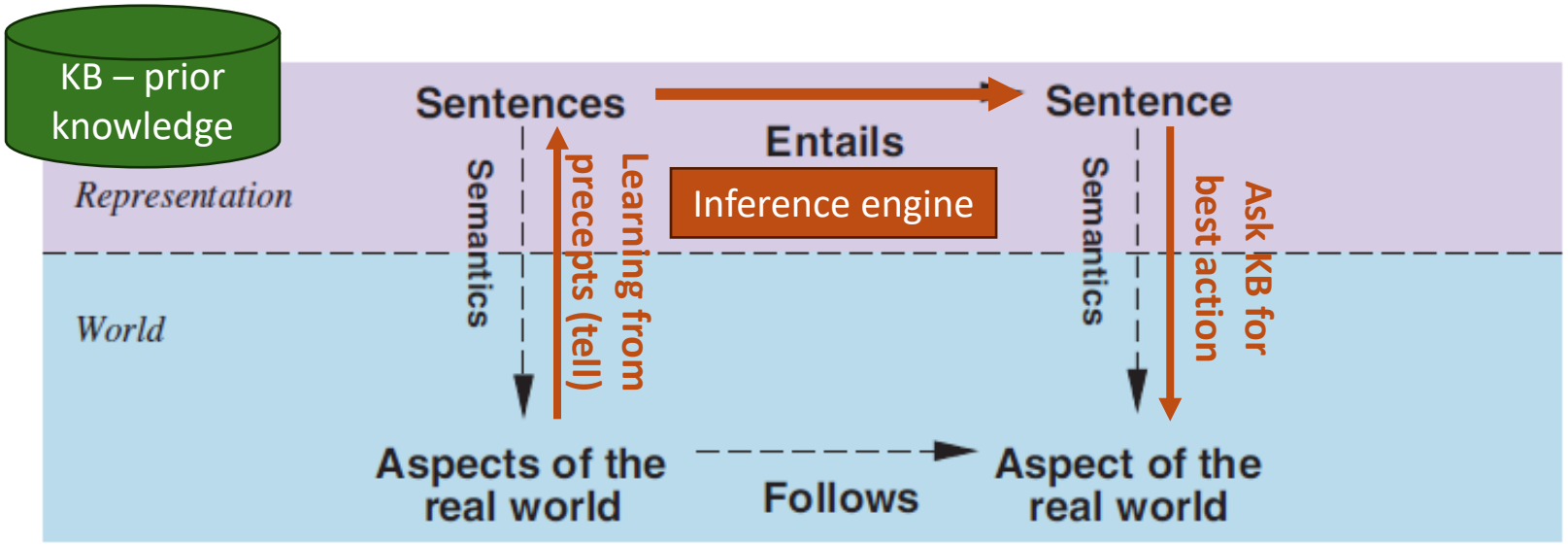
- **Facts:** Sentences we know to be true.
- **Possible worlds**: all worlds/models which are consistent with the facts we know (compare with belief state).
- **Learning** new facts reduces the number of possible worlds.
- **Entailment:** A new sentence logically follows from what we already know.
- **Reasoning**: The agent can reason about what will happen in the real world.

Knowledge-Based Agents



- Knowledge base (KB) = **set of facts.** E.g., set of **sentences** in a **formal language** that are known to be true.
- **Separation** between data (knowledge) and program (inference).
- **Declarative** approach to building an agent: Define what it needs to know in its KB. Use an off-the-shelf inference engine.
- The concepts of **goals or utility** can be incorporated into the knowledge base as facts.
- **Actions** are based on knowledge (sentences + inferred sentences).

Generic Knowledge-based Agent



function KB-AGENT(*percept*) **returns** an *action*
  persistent: KB, a knowledge base
    t, a counter, initially 0, indicating time

  TELL(KB, MAKE-PERCEPT-SENTENCE(*percept*, t))
  action $\leftarrow$ ASK(KB, MAKE-ACTION-QUERY(t))
  TELL(KB, MAKE-ACTION-SENTENCE(*action*, t))
  $t \leftarrow t + 1$
  return *action*

Memorize percept at time t

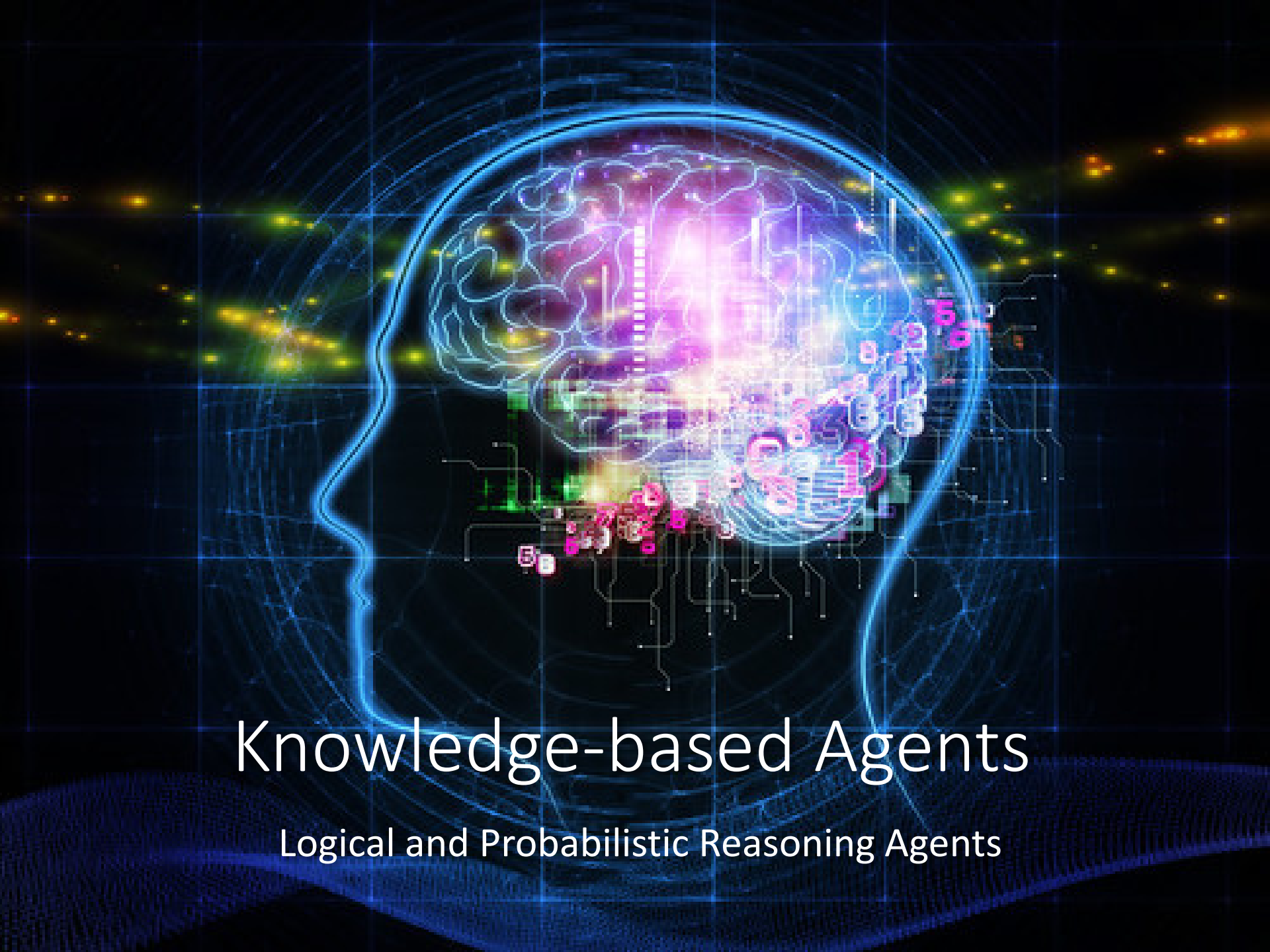
Ask for logical action given the knowledge

Record action taken at time t

Different Languages to Represent Knowledge

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

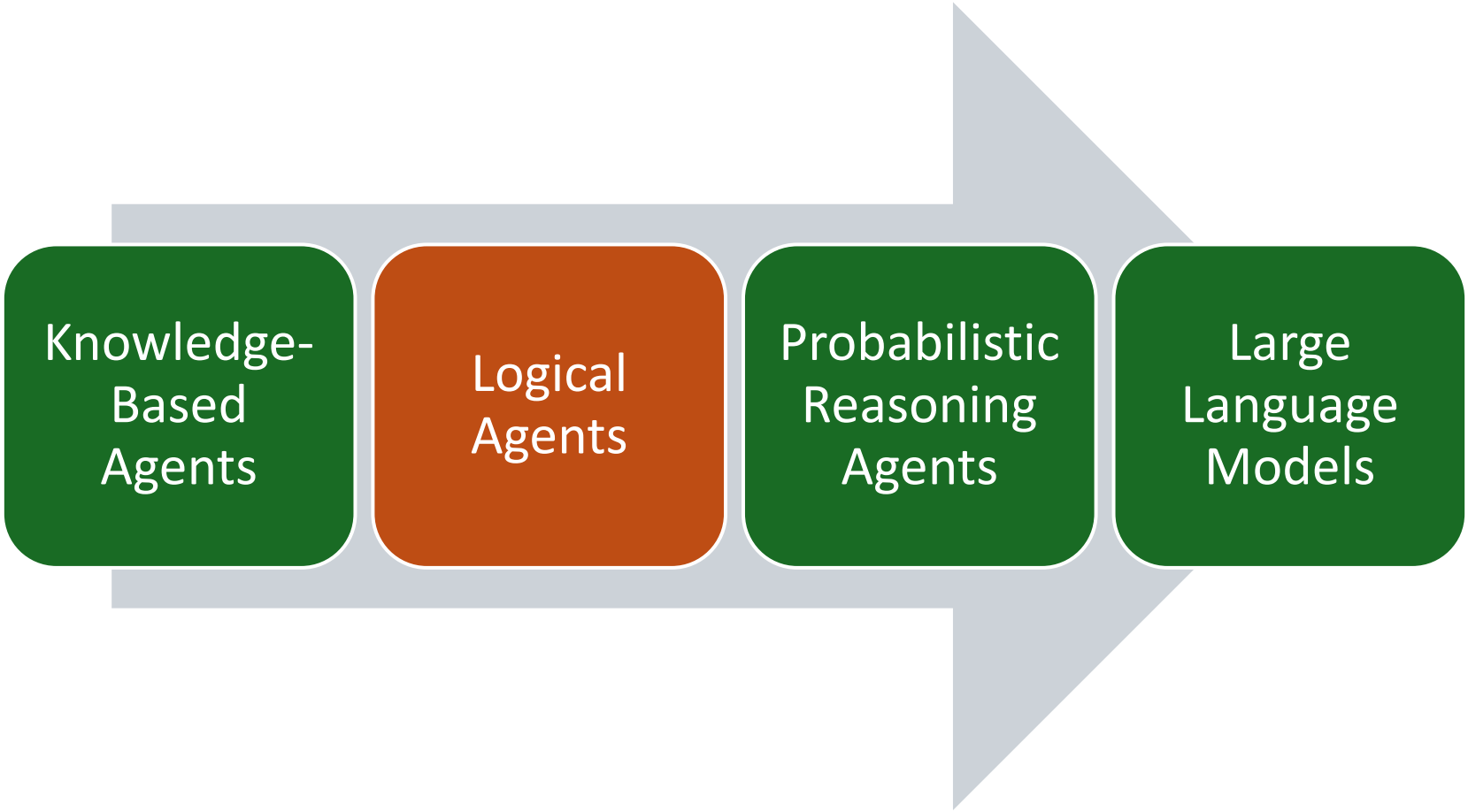
+ Natural Language — word patterns representing facts, objects, relations, … — ???



Knowledge-based Agents

Logical and Probabilistic Reasoning Agents

Outline


Knowledge-Based Agents
Logical Agents
Probabilistic Reasoning Agents
Large Language Models

Logical Agents

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

- Facts are logical sentences that are known to be true.
- Inference: Generate new sentences that are entailed by all known sentences.
- Implementation: Typically using Prolog
  - Declarative logic programing language.
  - Runs queries over the program (= the knowledge base)
- Synonyms: Symbolic AI, Expert Systems

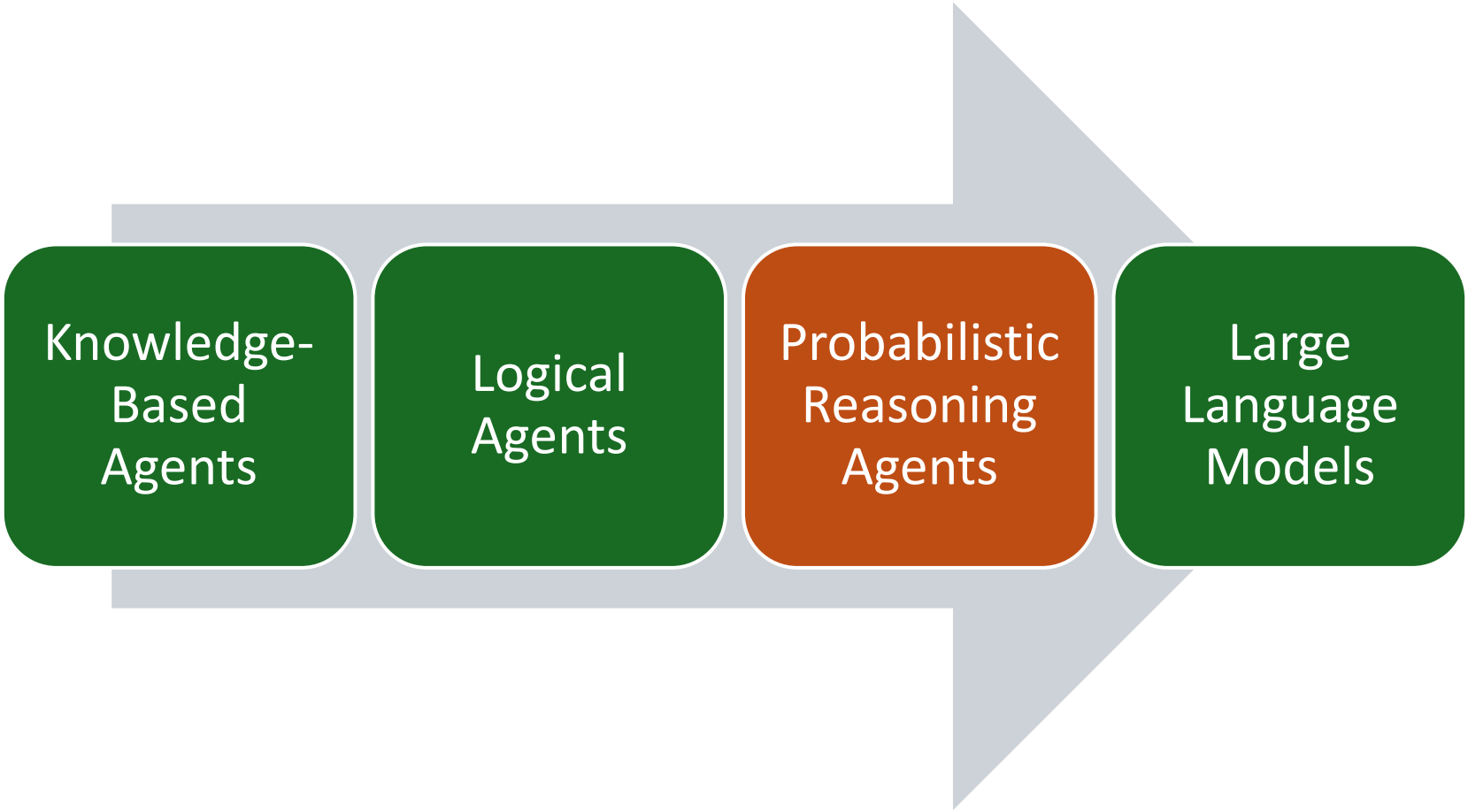
Issues:

- Inference is computationally very expensive.
- Logic cannot deal with uncertainty.

Code example:
A Simple Prolog Knowledge Base
The n-Queens Problem as a Logic Problem

See Code on https://michael.hahsler.net/Introduction_to_Artificial_Intelligence/

Outline


Knowledge-Based Agents
Logical Agents
Probabilistic Reasoning Agents
Large Language Models

Probabilistic Reasoning

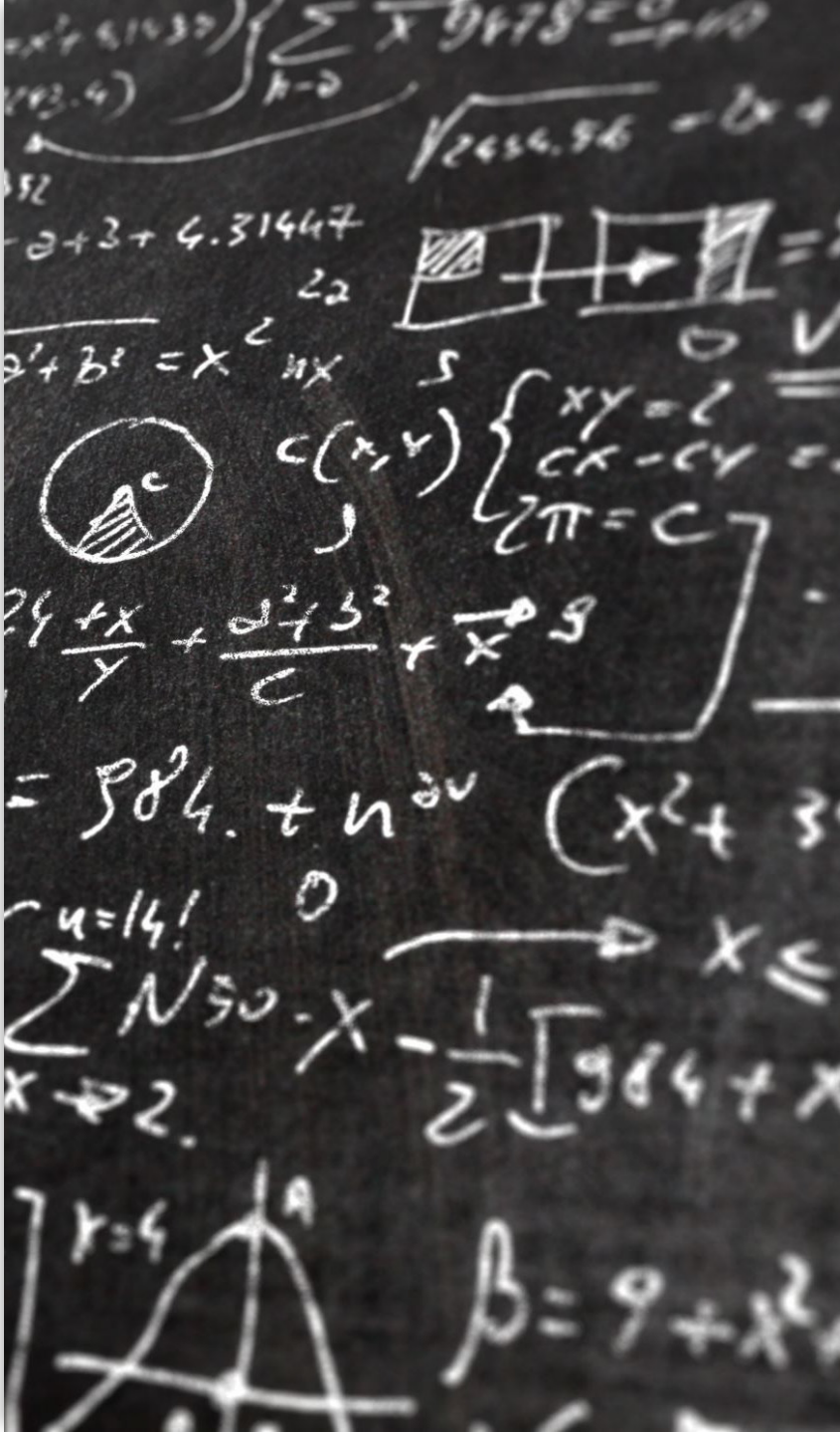
Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

- Replaces true/false with a probability.
- This is the basis for
  - Probabilistic reasoning under uncertainty
  - Decision theory
  - Machine Learning

We will talk about these topics a lot more

Conclusion

- The **clear separation between knowledge and inference engine** is very useful.
- **Pure logic** is often not flexible enough. The fullest realization of knowledge-based agents using logic was in the field of expert systems or knowledge-based systems in the 1970s and 1980s.
- **Pretrained Large Language Models** are an interesting new application of knowledge-based agents based on natural language.
- In one of the following chapters, we will talk about **probability theory,** the standard language to reason under uncertainty, and forms the basis of machine learning.

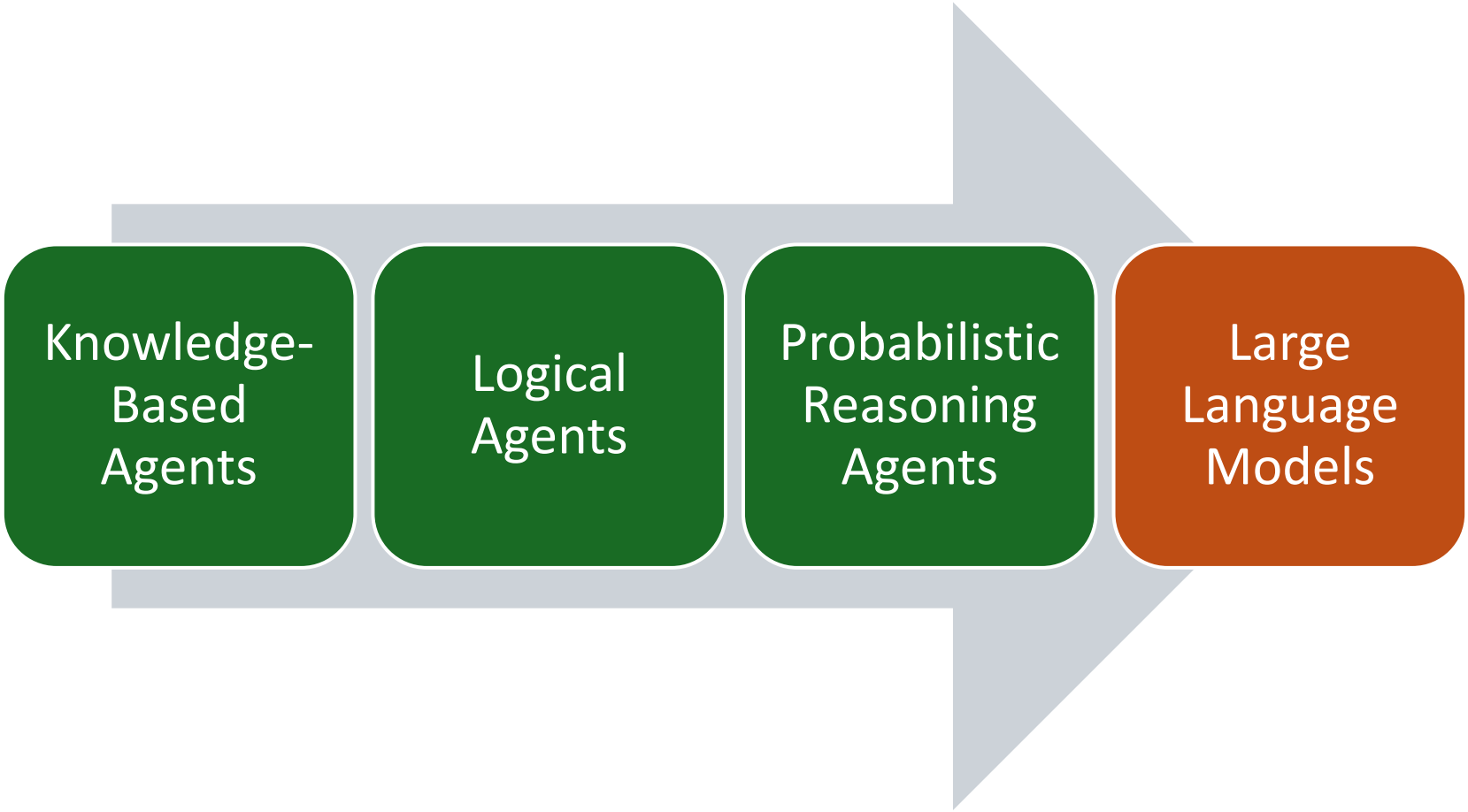




Knowledge-Based Agents using Natural Language

Large Language Models

Outline


Knowledge-Based Agents
Logical Agents
Probabilistic Reasoning Agents
Large Language Models

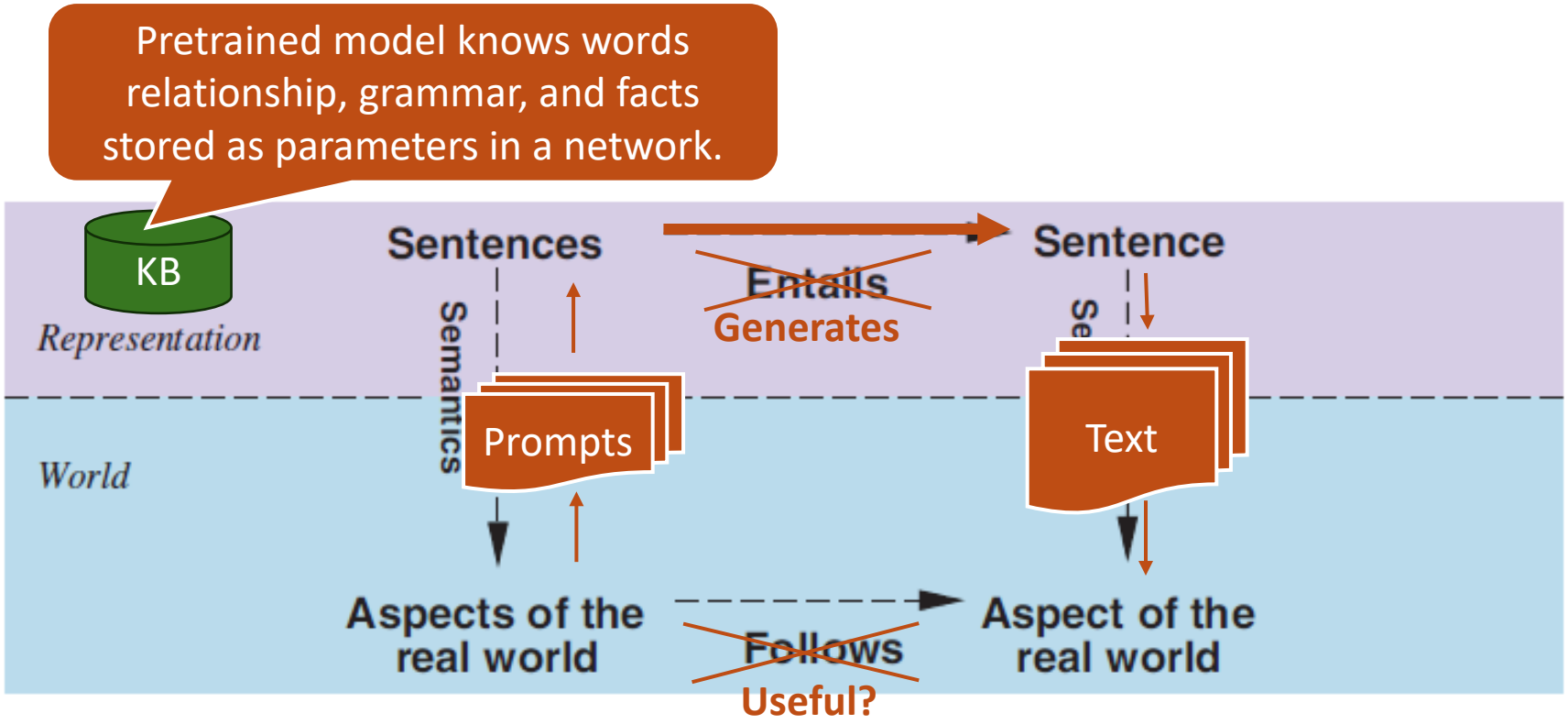
LLMs - Large Language Models

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

+	Natural Language	word patterns representing facts, objects, relations, …	???

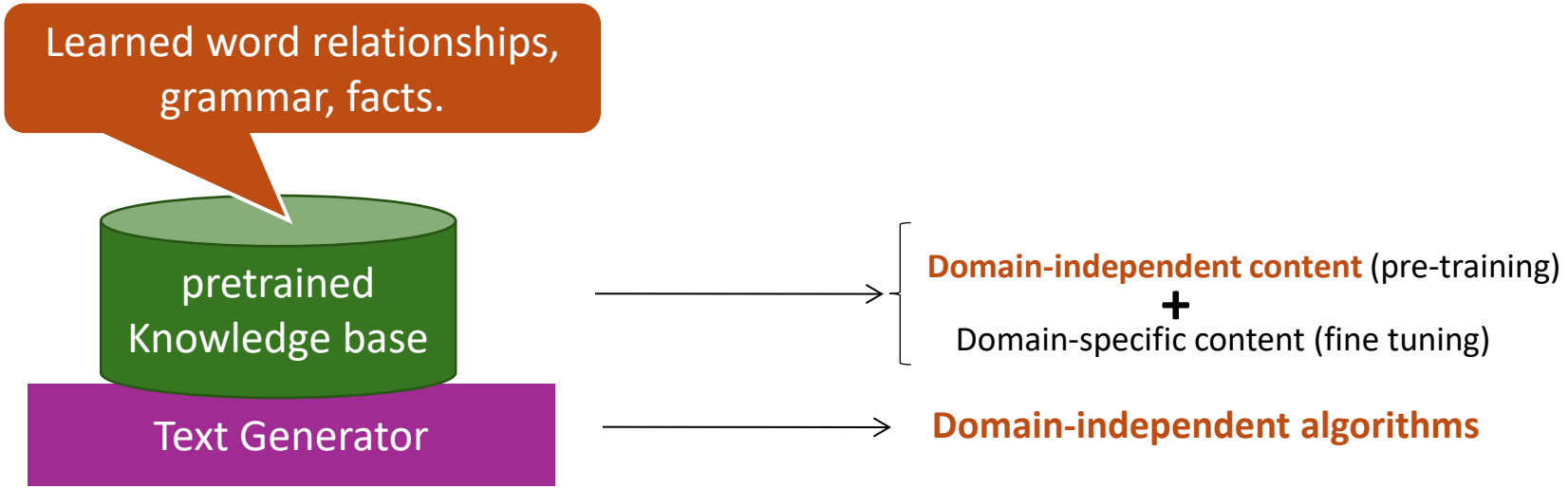
- Extract knowledge from large text corpora.
- Store knowledge compressed as parameters in a deep neural network.

Using Natural Language for Knowledge Representation



- The user formulates a question about the real world as a natural language **prompt** (a sequence of tokens).
- The LLM **generates text** using a pretrained model that represents its knowledge base.
- The text (hopefully) is useful in the real world. The **objective function** is not clear. Maybe it is implied in the prompt?

LLM as a Knowledge-Based Agents



Current text generators are:

- Pretrained decoder-only transformer models (e.g., GPT stands for Generative Pre-trained Transformer). The knowledge base is not updated during interactions.
- Tokens are created autoregressively. One token is generated at a time based on all the previous tokens using the transformer attention mechanism.

LLM as a Generic Knowledge-based Agent

Prompt + already generated tokens

function KB-AGENT(*percept*) **returns** an *action*
  persistent: KB, a knowledge base
    t, a counter, initially 0, indicating time

  ~~TELL(KB, MAKE-PERCEPT-SENTENCE(*percept*, t))~~
  $action \leftarrow$ ASK(KB, MAKE-ACTION-QUERY(t))
  ~~TELL(KB, MAKE-ACTION-SENTENCE(*action*, t))~~
  $t \leftarrow t + 1$
  return *action*

Next token

- A chatbot repeatedly calls the agent function till the agent function returns the ‘end’ token.

Code example:
Large Language Models

See Code on https://michael.hahsler.net/Introduction_to_Artificial_Intelligence/

Many Open Questions about LLMs

- Correlation is not causation: **Can LLMs reason** to solve problems?
- Leaky data makes it hard to evaluate true **reasoning performance**.
- Generative stochasticity leads to **hallucinations**: LLM makes up facts.
- Autoregression is an exponentially **diverging** diffusion process.
- The training data contains **biases**, nonsense, and harmful content.
- **Security**: LLM can leak sensitive information it was trained on.
- **Rights laundering:** Copyrighted or licensed material can be included in the training data.

Reading: [2307.04821] Amplifying Limitations, Harms and Risks of Large Language Models (arxiv.org)





Appendix: Logic

Details on Propositional and First-Order Logic

Logic to Represent Knowledge



Logic is a formal system for representing and manipulating facts (i.e., knowledge) so that true conclusions may be drawn



Syntax: rules for constructing valid sentences

E.g., $x + 2 \geq y$ is a valid arithmetic sentence, $\geq x2y +$ is not



Semantics: "meaning" of sentences, or relationship between logical sentences and the real world

Specifically, semantics defines truth of sentences

E.g., $x + 2 \geq y$ is true in a world where
$x = 5$
and $y = 7$

Propositional Logic

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

Propositional Logic: Syntax in Backus-Naur Form

$$
\begin{array}{rcll}
\mathit{Sentence} & \rightarrow & \mathit{AtomicSentence} \mid \mathit{ComplexSentence} & \\
\mathit{AtomicSentence} & \rightarrow & \mathit{True} \mid \mathit{False} \mid P \mid Q \mid R \mid \ldots & \text{= Symbols} \\
\mathit{ComplexSentence} & \rightarrow & (\ \mathit{Sentence}\ ) & \\
 & \mid & \neg\ \mathit{Sentence} & \text{Negation} \\
 & \mid & \mathit{Sentence} \wedge \mathit{Sentence} & \text{Conjunction} \\
 & \mid & \mathit{Sentence} \vee \mathit{Sentence} & \text{Disjunction} \\
 & \mid & \mathit{Sentence} \Rightarrow \mathit{Sentence} & \text{Implication} \\
 & \mid & \mathit{Sentence} \Leftrightarrow \mathit{Sentence} & \text{Biconditional}
\end{array}
$$

$$
\text{OPERATOR PRECEDENCE} \quad : \quad \neg, \wedge, \vee, \Rightarrow, \Leftrightarrow
$$

Validity and Satisfiability

A sentence is **valid** if it is true in **all** models/worlds

e.g., *True*, $A \vee \neg A$, $A \Rightarrow A$, $(A \wedge (A \Rightarrow B)) \Rightarrow B$
are called tautologies and are useful to deduct new sentences.

A sentence is **satisfiable** if it is true in **some** model

e.g., $A \vee B$, C
useful to find new facts that satisfy all current possible worlds.

A sentence is **unsatisfiable** if it is true in no models

e.g., $A \wedge \neg A$

Possible Worlds, Models and Truth Tables

A **model** specifies a "possible world" with the true/false status of each proposition symbol in the knowledge base

- E.g., **P** is true and **Q** is true
- With two symbols, there are $2^2 = 4$ possible worlds/models, and they can be enumerated exhaustively using:

A **truth table** specifies the truth value of a composite sentence for each possible assignments of truth values to its atoms. Each row is a model.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
$false$	$false$	$true$	$false$	$false$	$true$	$true$
$false$	$true$	$true$	$false$	$true$	$true$	$false$
$true$	$false$	$false$	$false$	$true$	$false$	$false$
$true$	$true$	$false$	$true$	$true$	$true$	$true$

We have 3 possible worlds for $P \Rightarrow Q = true$

Propositional Logic: Semantics

Rules for evaluating truth with respect to a model:

- $\neg$**P** is true iff **P** is false
- **P** $\wedge$ **Q** is true iff **P** is true and **Q** is true
- **P** $\vee$ **Q** is true iff **P** is true or **Q** is true
- **P** $\Rightarrow$ **Q** is true iff **P** is false or **Q** is true

Sentence

Model

Logical Equivalence

Two sentences are logically equivalent iff (read if, and only if) they are true in same models

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{de Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{de Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Entailment

- **Entailment** means that a sentence follows from the premises contained in the knowledge base:

$$\boldsymbol{KB} \models \boldsymbol{\alpha}$$

- The knowledge base *KB* entails sentence α iff α is true in all models where *KB* is true
  - E.g., KB with x = 0 entails sentence x * y = 0
- Tests for entailment
  - $\boldsymbol{KB} \models \boldsymbol{\alpha}$ iff $(\boldsymbol{KB} \Rightarrow \boldsymbol{\alpha})$ is *valid*
  - $\boldsymbol{KB} \models \boldsymbol{\alpha}$ iff $(\boldsymbol{KB} \wedge \neg \boldsymbol{\alpha})$ is *unsatisfiable*

Inference

- **Logical inference:** a procedure for generating sentences that follow from (ar entailed by) a knowledge base KB.

- An inference procedure is **sound** if it derives a sentence α iff KB ⊨ α. I.e, it only derives **true sentences**.

- An inference procedure is **complete** if it can derive **all** α for which *KB* ⊨ α.

Inference

- How can we check whether a sentence α is entailed by KB?

- How about we **enumerate all possible models of the KB** (truth assignments of all its symbols), and check that α is true in every model in which KB is true?
  - This is sound: All produced answer are correct.
  - This is complete: It will produce all correct answers.
  - **Problem**: if KB contains *n* symbols, the truth table will be of size 2^n

- Better idea: use ***inference rules***, or sound procedures to generate new sentences or *conclusions* given the *premises* in the KB.
- Look at the textbook for inference rules and resolution.

Inference Rules

- Modus Ponens

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

premises

conclusion

This means: If the KB contains the sentences $\alpha \Rightarrow \beta$ and α then β is true.

- And-elimination

$$\frac{\alpha \wedge \beta}{\alpha}$$

Inference Rules

- And-introduction

$$\frac{\alpha,\ \beta}{\alpha \wedge \beta}$$

- Or-introduction

$$\frac{\alpha}{\alpha \vee \beta}$$

Inference Rules

- Double negative elimination

$$\frac{\neg\neg\alpha}{\alpha}$$

- Unit resolution

$$\frac{\alpha \vee \beta, \neg\beta}{\alpha}$$

Resolution

$$\frac{\alpha \vee \beta, \neg\beta \vee \gamma}{\alpha \vee \gamma} \quad \text{or} \quad \frac{\alpha \vee \beta, \beta \Rightarrow \gamma}{\alpha \vee \gamma}$$

- Example:
  - α: “The weather is dry”
  - β: “The weather is rainy”
  - γ: “I carry an umbrella”

Resolution is Complete

$$\frac{\alpha \vee \beta, \neg\beta \vee \gamma}{\alpha \vee \gamma}$$

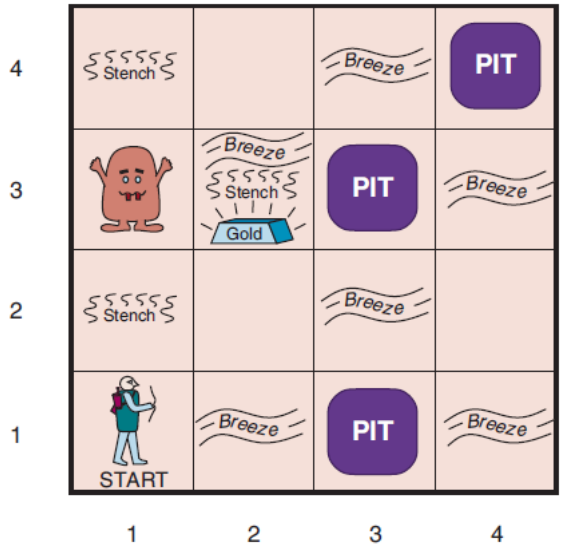
- To prove KB $\models \alpha$, assume KB $\wedge \neg \alpha$ and derive a contradiction
- Rewrite KB $\wedge \neg \alpha$ as a conjunction of *clauses*, or disjunctions of *literals*
  - *Conjunctive normal form* (CNF)
- Keep applying resolution to clauses that contain *complementary literals* and adding resulting clauses to the list
  - If there are no new clauses to be added, then KB does not entail α
  - If two clauses resolve to form an *empty clause*, we have a contradiction and KB $\models \alpha$

Complexity of Inference

- Propositional inference is ***co-NP-complete***
  - *Complement* of the SAT problem: $\alpha \not\models \beta$ if and only if the sentence $\alpha \wedge \neg \beta$ is *unsatisfiable*
  - Every known inference algorithm has worst-case exponential run time complexity.

- Efficient inference is only possible for restricted cases
  - e.g., Horn clauses are disjunctions of literals with at most one positive literal.

Example: Wumpus World



1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2	3,2	4,2
1,1 A OK	2,1 OK	3,1	4,1

(a)

A = *Agent*
B = *Breeze*
G = *Glitter, Gold*
OK = *Safe square*
P = *Pit*
S = *Stench*
V = *Visited*
W = *Wumpus*

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

(b)

Example: Wumpus World

Initial KB needs to contain rules like these for each square:

$Breeze(1,1) \Leftrightarrow Pit(1,2) \vee Pit(2,1)$
$Breeze(1,2) \Leftrightarrow Pit(1,1) \vee Pit(1,3) \vee Pit(2,2)$
$\text{Stench}(1,1) \Leftrightarrow W(1,2) \vee W(2,1)$
…

We have to enumerate all possible scenarios in propositional logic! First-order logic can help.

Percepts at (1,1) are no breeze or stench. Add the following facts to the KB:

$\neg Breeze(1,1)$
$\neg Stench(1,1)$

Inference will tell us that the following facts are entailed:

$\neg Pit(1,2), \neg Pit(2,1), \neg W(1,2), \neg W(2,1)$

This means that $(1,2)$ and $(2,1)$ are safe.

Summary

- Logical agents apply inference to a knowledge base to derive new information and make decisions.
- Basic concepts of logic:
  - syntax: formal structure of sentences
  - semantics: truth of sentences in models
  - entailment: necessary truth of one sentence given another
  - inference: deriving sentences from other sentences
  - soundness: derivations produce only entailed sentences
  - completeness: derivations can produce all entailed sentences
- Resolution is complete for propositional logic.
- Algorithms use forward, backward chaining, are linear in time, and complete for special clauses (definite clauses).

Limitations of Propositional Logic

Suppose you want to say “All humans are mortal”

- In propositional logic, you would need ~6.7 billion statements of the form:

  Michael_Is_Human and Michael_Is_Mortal,

  Sarah_Is_Human and Sarah_Is_Mortal, …

Suppose you want to say “Some people can run a marathon”

- You would need a disjunction of ~6.7 billion statements:

  Michael_Can_Run_A_Marathon or … or Sarah_Can_Run_A_Marathon

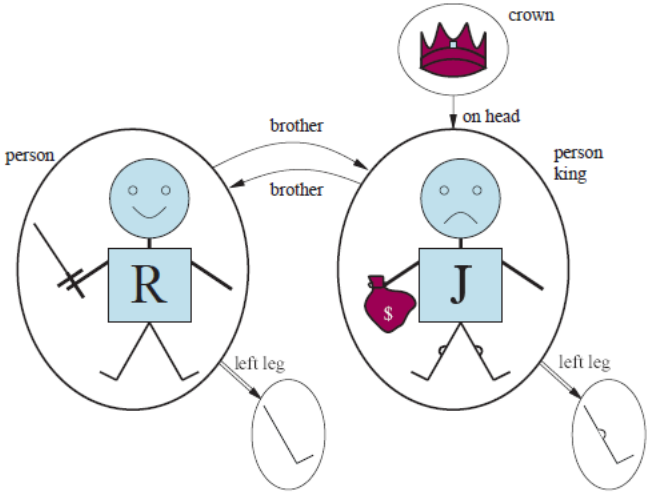
First-Order Logic

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

First-order Logic adds **objects** and **relations** to the facts of propositional logic.

This addresses the issues of propositional logic, which needs to store a fact for each instance of and object individually.

Syntax of FOL



$$
\begin{aligned}
\textit{Sentence} &\rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence} \\
\textit{AtomicSentence} &\rightarrow \textit{Predicate} \mid \textit{Predicate}(\textit{Term}, \ldots) \mid \textit{Term} = \textit{Term} \\
\textit{ComplexSentence} &\rightarrow (\ \textit{Sentence}\ ) \\
&\mid \neg\ \textit{Sentence} \\
&\mid \textit{Sentence} \wedge \textit{Sentence} \\
&\mid \textit{Sentence} \vee \textit{Sentence} \\
&\mid \textit{Sentence} \Rightarrow \textit{Sentence} \\
&\mid \textit{Sentence} \Leftrightarrow \textit{Sentence} \\
&\mid \textit{Quantifier}\ \textit{Variable}, \ldots\ \textit{Sentence} \\
\\
\textit{Term} &\rightarrow \textit{Function}(\textit{Term}, \ldots) \\
&\mid \textit{Constant} \\
&\mid \textit{Variable} \\
\\
\textit{Quantifier} &\rightarrow \forall \mid \exists \\
\textit{Constant} &\rightarrow A \mid X_1 \mid \textit{John} \mid \cdots \\
\textit{Variable} &\rightarrow a \mid x \mid s \mid \cdots \\
\textit{Predicate} &\rightarrow \textit{True} \mid \textit{False} \mid \textit{After} \mid \textit{Loves} \mid \textit{Raining} \mid \cdots \\
\textit{Function} &\rightarrow \textit{Mother} \mid \textit{LeftLeg} \mid \cdots \\
\text{OPERATOR PRECEDENCE} &: \neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow
\end{aligned}
$$

Objects

Relations. Predicate is/returns True or False

Function returns an object

Universal Quantification

- $\forall$**x P(x)**
- Example: "Everyone at SMU is smart"
  $\forall$**x At(x,SMU)** $\Rightarrow$ **Smart(x)**
  Why not $\forall$**x At(x,SMU)** $\wedge$ **Smart(x)**?
- Roughly speaking, equivalent to the conjunction of all possible instantiations of the variable:
  [At(John, SMU) $\Rightarrow$ **Smart(John)]** $\wedge$ **...**
  [At(Richard, SMU) $\Rightarrow$ **Smart(Richard)]** $\wedge$ **...**
- $\forall$**x P(x)** is true in a model m iff **P(x)** is true with **x** being each possible object in the model

Existential Quantification

- **$\exists$x P(x)**

- Example: "Someone at SMU is smart"
  $\exists$x At(x,SMU) $\wedge$ Smart(x)
  Why not **$\exists$x At(x,SMU) $\Rightarrow$ Smart(x)**?

- Roughly speaking, equivalent to the disjunction of all possible instantiations:
  [At(John,SMU) $\wedge$ Smart(John)] $\vee$
  [At(Richard,SMU) $\wedge$ Smart(Richard)] $\vee$ **…**

- **$\exists$x P(x)** is true in a model m iff **P(x)** is true with **x** being some possible object in the model

Properties of Quantifiers

- ∀**x** ∀**y** is the same as ∀**y** ∀**x**
- ∃**x** ∃**y** is the same as ∃**y** ∃**x**
- ∃**x** ∀**y** is not the same as ∀**y** ∃**x**

  ∃**x** ∀**y Loves(x,y)**

  "There is a person who loves everyone"

  ∀**y** ∃**x Loves(x,y)**

  "Everyone is loved by at least one person"

- **Quantifier duality:** each quantifier can be expressed using the other with the help of negation

  ∀**x Likes(x,IceCream)** ¬∃**x**

  ∃**x Likes(x,Broccoli)** ¬∀**›**

Equality

- **$\text{Term}_1 = \text{Term}_2$** is true under a given model if and only if **Term_1** and **Term_2** refer to the same object

- E.g., definition of **Sibling** in terms of **Parent**:

  $\forall$**x,y Sibling(x,y)** $\Leftrightarrow$
  [$\neg$(x = y) $\wedge$ $\exists$**m,f** $\neg$ **(m = f)** $\wedge$ **Parent(m,x)** $\wedge$ **Parent(f,x)** $\wedge$
  Parent(m,y) $\wedge$ **Parent(f,y)]**

Example: The Kinship Domain

- Brothers are siblings

  $\forall$**x,y Brother(x,y)** $\Rightarrow$ **Sibling(x,y)**

- "Sibling" is symmetric

  $\forall$**x,y Sibling(x,y)** $\Leftrightarrow$ **Sibling(y,x)**

- One's mother is one's female parent

  $\forall$**m,c (Mother(c) = m)** $\Leftrightarrow$ **(Female(m)** $\wedge$ **Parent(m,c))**

Example: The Set Domain

- $\forall s\ \mathbf{Set}(s) \Leftrightarrow (s = \{\}) \vee (\exists x, s_2\ \mathbf{Set}(s_2) \wedge s = \{x \mid s_2\})$
- $\neg\exists x, s\ \{x \mid s\} = \{\}$
- $\forall x, s\ x \in s \Leftrightarrow s = \{x \mid s\}$
- $\forall x, s\ x \in s \Leftrightarrow [\exists y, s_2\ (s = \{y \mid s_2\} \wedge (x = y \vee x \in s_2))]$
- $\forall s_1, s_2\ s_1 \subseteq s_2 \Leftrightarrow (\forall x\ x \in s_1 \Rightarrow x \in s_2)$
- $\forall s_1, s_2\ (s_1 = s_2) \Leftrightarrow (s_1 \subseteq s_2 \wedge s_2 \subseteq s_1)$
- $\forall x, s_1, s_2\ x \in (s_1 \cap s_2) \Leftrightarrow (x \in s_1 \wedge x \in s_2)$
- $\forall x, s_1, s_2\ x \in (s_1 \cup s_2) \Leftrightarrow (x \in s_1 \vee x \in s_2)$

Inference in FOL

Inference in FOL is complicated!

1. **Reduction to propositional logic** and then use propositional logic inference.

2. **Directly do inference on FOL (or a subset like definite clauses)**
   - Unification: Combine two sentences into one.
   - Forward Chaining for FOL
   - Backward Chaining for FOL
   - Logical programming (e.g., Prolog)